

KnowSelf

Teaching LLM Agents When to Think,
Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness: regulate reflection and
retrieval by situation

Source: arxiv.org/abs/2504.03553

'Flood irrigation' in agent learning

Blindly inject trajectories/feedback/knowledge regardless of situational need

Trajectory imitation

e.g., ReAct

- Learns patterns from gold trajectories
- Can collapse on unexpected inputs

Trial-and-error refinement

e.g., Reflexion

- Injects external feedback by default
- Extra compute even when not needed

Knowledge-augmented planning

e.g., KnowAgent / WKM

- Queries knowledge at (nearly) every step
- High inference cost; can still be brittle

Source: <https://arxiv.org/abs/2504.03553> (problem framing)

Why it matters

Humans assess the situation:

- act directly when confident
- pause to reflect when uncertain
- seek external knowledge when stuck

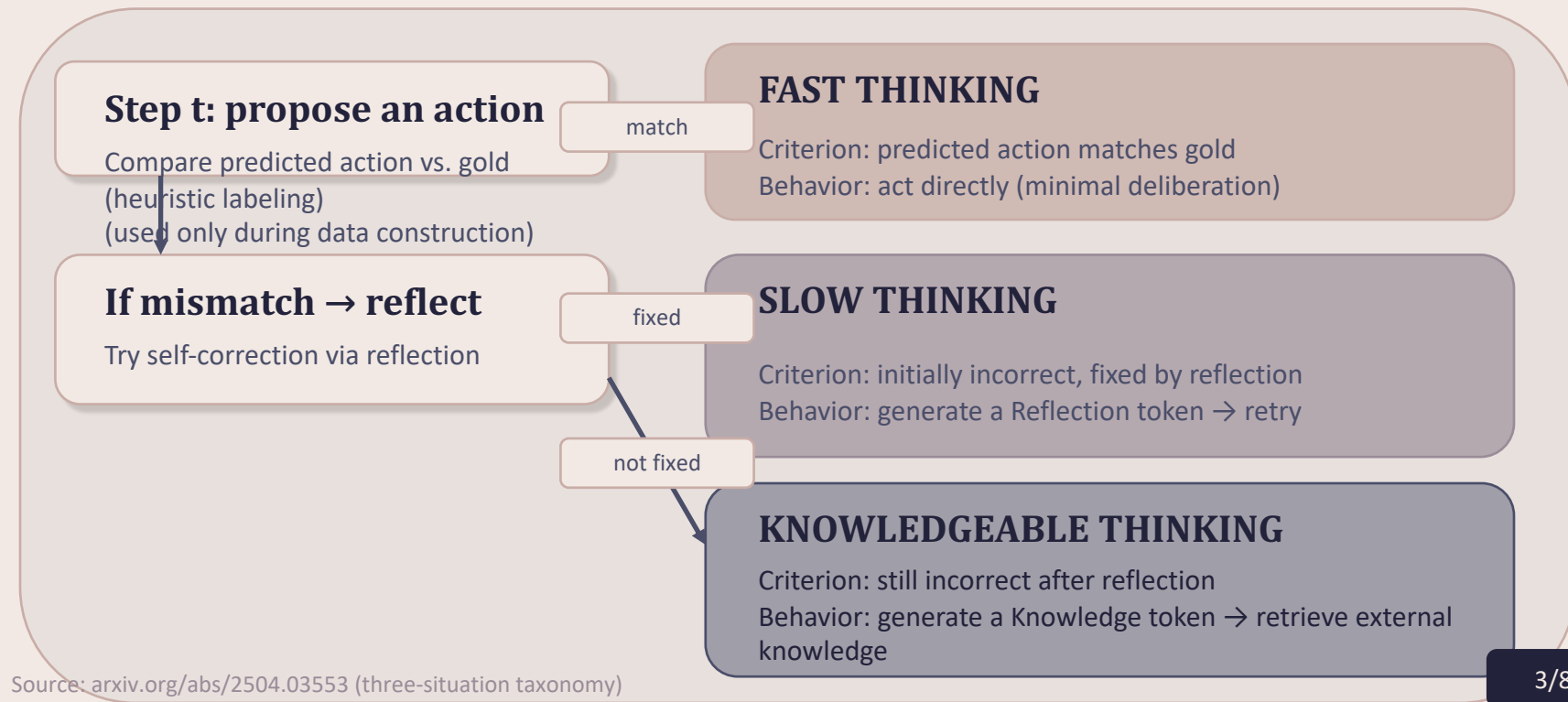
Flood irrigation leads to:

- fragile planning
- pattern collapse on unexpected inputs
- unnecessarily high inference cost

Goal: situational self-awareness

Three situational thinking modes

Decision-time self-awareness: act → reflect → retrieve (if needed)



KnowSelf pipeline: construct → train → infer

Special tokens encode situational self-awareness and trigger compute/knowledge only when needed

1

Self-awareness data construction

Self-explore trajectories; heuristic labels each step:

- match gold → Fast
- reflection fixes → Slow

2

Two-stage training

SFT learns initial token patterns; RPO refines when to emit special tokens.

3

Self-aware inference

At runtime, model emits tokens that route execution:

- act directly
- reflect
- retrieve external knowledge

Framework figure

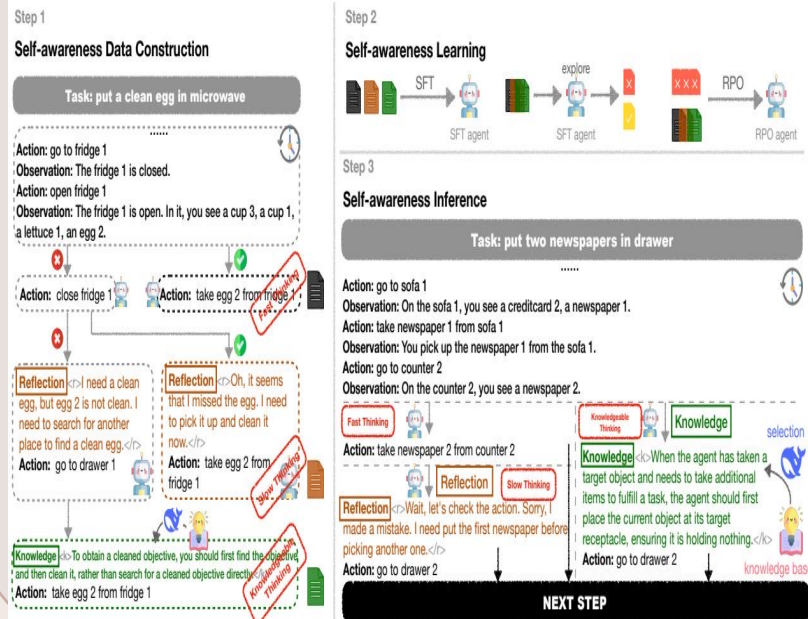


Figure: KnowSelf framework (figure_3) from arxiv.org/abs/2504.03553

Qualitative grounding: reflection vs. knowledge

KnowSelf triggers the right mode only when needed

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

Task A: “put a hot mug in cabinet”

Figure 1: KnowSelf (Llama-8B) (figure 1: Reflection in [table 2](#))

KnowSelf (Llama-8B) uses a Reflection token to realize it must heat the mug using the open microwave:

Task B: “put a saltshaker in drawer”

KnowSelf applies a Knowledge token: place the object directly without removing unrelated items: DeepSeek

ALFWorld success rates (%) + knowledge usage

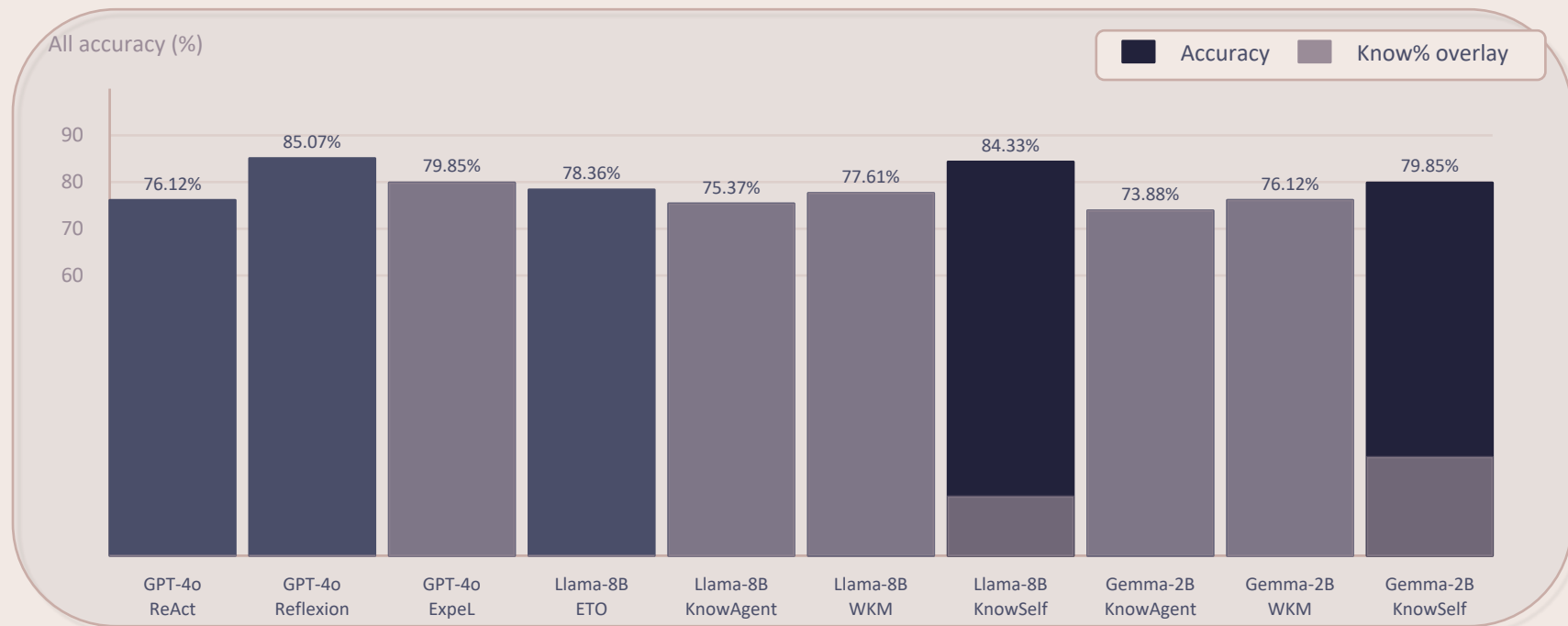
KnowSelf wins with selective retrieval (Know% ≈ 15–26%)

Backbone	Method	Know%	Put	Clean	Heat	Cool	Exam.	Put2	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Key highlights: Llama-8B KnowSelf = 84.33% @ 15.01% Know, Gemma-2B KnowSelf = 79.85% @ 26.41% Know
 Data: ALFWorld results table from arxiv.org/abs/2504.03553 (see source_brief.md)

Efficiency: knowledge usage vs. overall success

KnowSelf achieves top accuracy with ~15–26% knowledge queries (vs. 100% baselines)



Interpretation: Full-knowledge methods (Know%=100%) do not guarantee higher success; KnowSelf learns when retrieval is actually necessary.

Data: All + Know% from ALFWorld table (source_brief.md)

Key takeaways

Self-awareness is the missing piece in agent planning

- Indiscriminate knowledge injection (“flood irrigation”) is wasteful and can be suboptimal.
- Situational self-awareness can be learned via special tokens + lightweight heuristic labeling on self-explored trajectories.
- KnowSelf (Llama-8B) reaches 84.33% on ALFWorld, outperforming GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).
- Only 15–26% knowledge usage is needed versus 100% for knowledge-heavy baselines (ExpeL / KnowAgent / WKM).
- Generalizes across model scales: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.